

In-Class Python Task Worksheet

Arrays and Algorithm Analysis - Tasks 1 to 3

Student Name: _____

Student ID: _____

Date: _____

Class/Section: _____

Instructions: Predict first, then run the code. Write the required Big-O complexity and a brief justification. Submit this worksheet before the end of class.

Task 1 - Complexity Challenge

Determine the **time complexity** of each code fragment and briefly explain why.

A.

```
print(numbers[0])
```

Time Complexity: _____ Reason: _____

B.

```
for i in range(n):  
    print(i)
```

Time Complexity: _____ Reason: _____

C.

```
for i in range(n):  
    for j in range(n):  
        print(i, j)
```

Time Complexity: _____ Reason: _____

D.

```
for i in range(n):  
    print(i)
```

```
for j in range(n):  
    print(j)
```

Time Complexity: _____ Reason: _____

E.

```
for i in range(n):  
    for j in range(100):  
        print(i, j)
```

Time Complexity: _____ Reason: _____

Task 2 - Logarithmic Complexity Challenge

Study the code below. Predict the printed sequence before running it, then analyze its time complexity.

```
n = 100

while n > 1:
    print(n)
    n = n // 2
```

1. Write the values you expect the program to print:

2. What happens to n after every loop iteration?

3. State the time complexity in Big-O notation:

Time Complexity: _____

4. Explain why this loop is not $O(n)$:

Extension: If the starting value were **1024** instead of 100, approximately how many times would the loop execute?

Answer: _____ Explanation: _____

Task 3 - Space Complexity

Analyze the **auxiliary space complexity** of each algorithm. Do not count the input list itself as additional memory.

Algorithm A

```
def calculate_sum(numbers):  
    total = 0  
  
    for number in numbers:  
        total += number  
  
    return total
```

A1. Auxiliary space complexity: _____

A2. Which variables use additional memory?

Algorithm B

```
def double_numbers(numbers):  
    result = []  
  
    for number in numbers:  
        result.append(number * 2)  
  
    return result
```

B1. Auxiliary space complexity: _____

B2. Why does the memory usage grow with the input size?

Comparison Question

Compare Algorithm A and Algorithm B. Which one uses less auxiliary memory, and why?

Submission Check: Predicted before running: Yes / No Tested in Python: Yes / No

Student signature: _____